Experimental Investigation of Harmonic Contributions to Spectral Filter Equivalent Widths

Bora Bozdag

November 2024

Abstract

Spectral filters are a key part of a powerful tool, remote temperature sensing. This method of measuring the temperature of an object by detecting the radiated electromagnetic waves is dependent on the accuracy of the optical equipment. This experiment investigated the variations in the widths of the transmission functions of spectral filters due to the contributions from the harmonics of the mean wavelength of that spectral filter. Investigation of the widths for three spectral filters provided evidence for the contributions from the harmonics to the widths being significant. In three further remote temperature sensing measurements on an identical window and a blackbody system, accuracies of ±2% and ±0.5% were obtained in window and blackbody temperature measurements, respectively. These results allowed confidence in the accuracy and the reliability of the setup used for measurements.

1 Introduction

Remote temperature sensing is a method that determines the brightness temperature of an object from a distance by measuring the electromagnetic waves radiated from it [1]. Temperature measurements of objects which are not accessible via local measurements are of major interest in many fields such as meteorology [2], oceanography [3, 4] and cosmology [5]. Layer-resolving temperature measurements require a number of spectral filters to distinguish the individual layers of the target (such as an atmosphere), which demands high accuracy in spectral filter characterisation.

The presence of the higher harmonics of the mean wavelength of a spectral filter causes a deviation in the transmission function from a Gaussian, which introduces significant errors for systems that cannot eliminate these harmonics via Fourier decomposition. This investigation studies the contributions of the higher harmonics to a spectral filter's equivalent width, which is the full width at half maximum (FWHM) of the spectral filter's transmittance function. Comparison of the filtered and unfiltered harmonic cases demonstrated an increase in the FWHM for two out of the three spectral filters tested, with the exception as an anomaly due to the spectral filter being defective. This concluded that there is a significant contribution to the equivalent widths from the harmonics of the mean wavelength. Further tests on an identical remote sensing setup justified the accuracy of the found results, validating the investigation on the change of equivalent width of spectral filters.

2 Theory

2.1 Spectral Filters

An ideal spectral filter has a transmission function $\tau(\lambda) = \delta(\lambda - \lambda_f)$ where λ_f is the mean wavelength of the filter and $\delta(\lambda)$ is a delta function. Non-ideal spectral filters deviate from the ideal case in two ways: Firstly, around $\lambda = \lambda_f$ the transmission function is a Gaussian and has an equivalent width, which is the FWHM of the Gaussian. This means that the filter transmits a range of wavelengths close to λ_f. Lastly, spectral filters also transmit higher harmonics of λ_f, which are represented by Gaussians at $\lambda = \lambda_f/n$ where $n \in Z$. These two features are presented in Figure 1a.

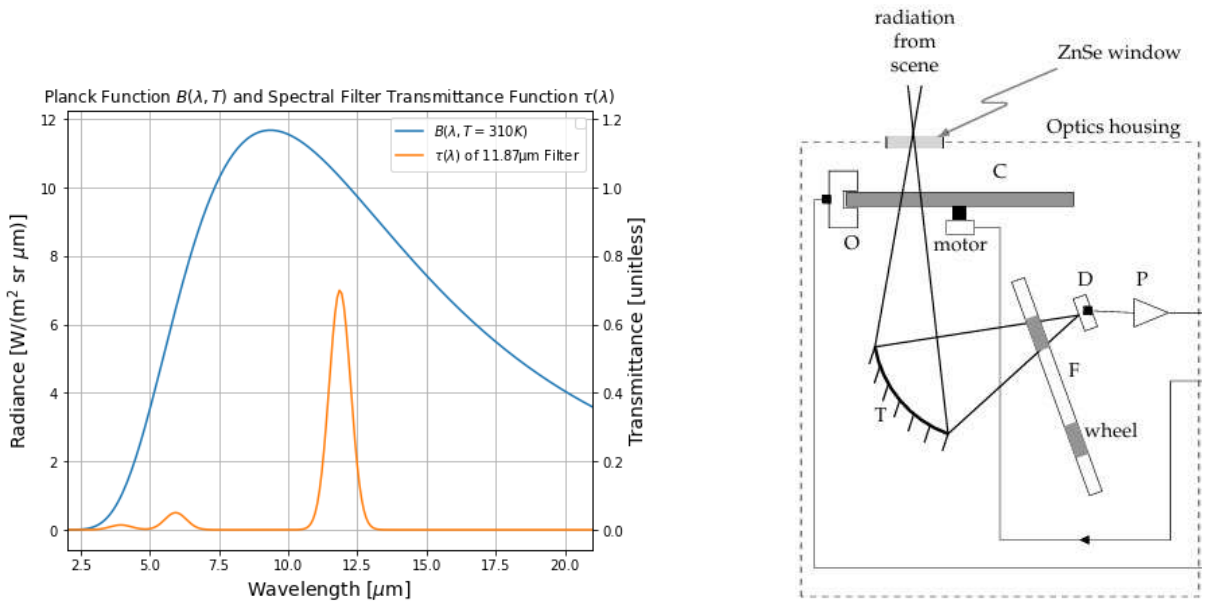
The radiance of a blackbody at temperature T, after passing through a spectral filter with transmittance $\tau_f(\lambda)$ is modelled using the Planck function, $B(\lambda, T)$ as defined in Appendix A.1, as [6]

$$R_f(T) = \int B(\lambda, T)\tau_f(\lambda)d\lambda \simeq B(\lambda_f, T)\int \tau_f(\lambda)d\lambda \equiv B(\lambda_f, T)\Delta\lambda_f \ . \tag{1}$$

The approximation used here assumes the Planck function to vary only linearly over the integral domain due to small equivalent widths (refer to Figure 1a).

2.2 Remote Sensing

Consider a system of a blackbody source of temperature T_p and a window of temperature T_w as the scene the radiometer is viewing. The incoming radiation to the window is transmitted $\propto \tau_w(\lambda)$, assuming the window to be a blackbody. The radiation not transmitted must be absorbed and re-emitted $\propto [1 - \tau_w(\lambda)]$.



(a) (b)

Figure 1: (a) Graph presenting the spectral filter transmittance function, $\tau(\lambda_f = 11.87\mu\text{m})$, and the Planck function, B(λ, $T = 310$K). (b) Diagram of the radiometer used in this experiment. C: chopper disc, T: curved mirror, F: spectral filter wheel, D: detector, P: pre-amplifier, O: phase sensor. Radiation enters through a ZnSe window. Adapted from [6].

Hence, the radiance measured from a window and a blackbody system through a spectral filter of mean wavelength λ_f is given by [6]

$$R_f = B(\lambda_f, T_p)\tau_w(\lambda_f) + B(\lambda_f, T_w)\left[1 - \tau_w(\lambda_f)\right] \ . \tag{2}$$

Spectral filters of different λ_f can be used to obtain a simultaneous set of equations of the form of Equation (2). The number of simultaneous equations depends on the number of windows/layers. Solving these equations obtains the brightness temperatures, T_i, of each part of the system. This experiment uses three spectral filters of mean wavelengths: $\lambda_f = 8.16\mu\text{m}, 10.62\mu\text{m}$ and $11.87\mu\text{m}$ respectively.

2.3 Signal Processing

The radiation entering the radiometer first passes through the chopper disc, C. This introduces a fixed frequency time variation to the radiation incident on the detector. The detector measures a signal, S, proportional to the input radiance as detailed in Appendix A.3. This signal, S, is the amplitude of the modulated component of the total signal measured by the detector, and is isolated by a synchronous rectifier. The total signal measured, $S_{total}(t)$, is given by [6]

$$S_{total}(t) = (S_e - S_C)\,F(t) + S_C + S_i \ , \tag{3}$$

$$S = S_e - S_C \ , \tag{4}$$

where S_e, S_i and S_C are the contributions from the external radiation, the internal components and the chopper disc respectively. S_i is not modulated as the chopper disc, C, is placed before the internal optical components – shown in Figure 1b. For this experiment, $F(t)$ is a 30 Hz square wave.

Given that the signal measured, S, is proportional to the input radiance, viewing two blackbodies at different temperatures without any spectral filters (through an open window) gives a signal difference [6]

$$\Delta S = \Delta S_w = a\left[\bar{B}(T_h) - \bar{B}(T_c)\right] = a\frac{\sigma}{\pi}\left(T_h^4 - T_c^4\right) \ , \tag{5}$$

where T_h and T_c are the hot and cold blackbody temperatures, a is a constant and σ is the Stefan-Boltzmann constant. The second equation follows from the Stefan-Boltzmann law and a detailed derivation is presented in Appendix A.1. However, when the spectral filters are replaced, we recover a signal difference

$$\Delta S = \Delta S_f = a \left[B\left(\lambda_f, T_h\right) - B\left(\lambda_f, T_c\right) \right] \Delta\lambda_f \ , \tag{6}$$

derived from Equation (1) for a signal S_f proportional to radiance R_f.

2.4 Transmittances

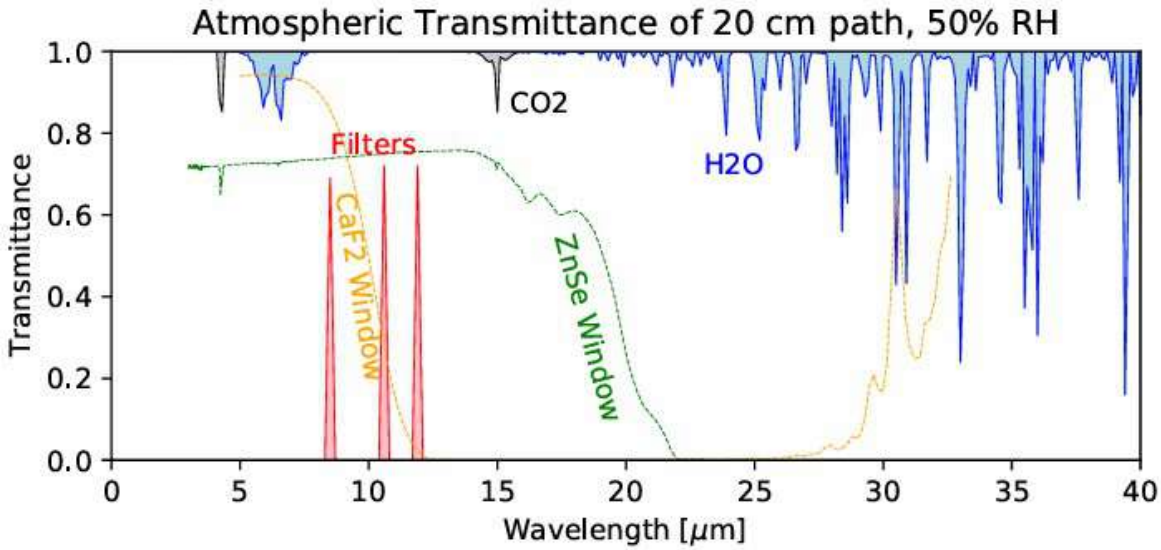


Figure 2: Transmittances of the spectral filters, the ZnSe and the CaF_2 windows, and the atmosphere: CO_2, H_2O. Reproduced from [6].

3 Methodology

3.1 Model

This experiment models the window-blackbody system explained in Section 2.2 as an analogue of a planetary body (blackbody source) and a single layer atmosphere (CaF_2 window), having temperatures T_p and T_w, respectively. This is what the radiometer views from its ZnSe window as the scene. Several past studies have used a similar model [7, 8].

3.2 Setup and Measurements

The components of the radiometer are shown in Figure 1b. On the spectral filter wheel, F, there are three spectral filters of mean wavelengths $\lambda_f = 8.16\mu\text{m}, 10.62\mu\text{m}$ and $11.87\mu\text{m}$, respectively and an open window. Radiation from the scene enters the radiometer through the ZnSe window. The window is chosen to filter out wavelengths above 20μm in order to work in the minimal atmospheric absorption region (refer to Figure 2). The detector, D, used measures the total intensity but is not able to use Fourier decomposition to distinguish different wavelengths. Instead, given that the peak at the mean wavelength, λ_f, of the transmission function dominates over the harmonic peaks (refer to Figure 1a), the transmission functions will be idealised to a single Gaussian for each spectral filter, centred at $\lambda = \lambda_f$, with an effective FWHM.

The first part of this experiment measures the signal difference between two blackbody sources of different and known temperatures. Repeating this for each spectral filter and the open window cases, Equations (5) and (6) are solved to calculate an effective $\Delta\lambda_f$ for each spectral filter. Then a thicker ZnSe window is introduced, which filters out the harmonics before they enter the radiometer, and another set of $\Delta\lambda_f$'s are calculated for each spectral filter. These calculations result in two transmission functions for each spectral filter, both centred at the respective mean wavelength, λ_f, of that spectral filter. But the first transmission function has a width of an effective $\Delta\lambda_f$ which has harmonic contributions, whereas the second transmission function has the actual width of the peak at $\lambda = \lambda_f$ as the harmonics are filtered out.

Furthermore, the expected input radiance, $R_f(T)$, curve is found using the Planck function and the calculated $\Delta\lambda_f$ values through Equation (1). This is a required check to determine the validity of the results. Physically, filtering out radiation cannot increase the input radiance, $R_f(T)$. For a non-anomalous result, the $R_f(T)$ curve for the thicker ZnSe window case must lie below the $R_f(T)$ curve for the normal ZnSe window case at all temperatures.

The second part of this experiment performs remote temperature sensing on the blackbody source and CaF_2 system discussed in Section 3.1. For each of the three spectral filters from the first experiment, a signal, S, is measured and converted to input radiance, R_f, using their proportionality (explained in Appendix A.3). These R_f values obtained three simultaneous equations of the form of Equation (2), which are then solved in pairs to get three pairs of T_p and T_w values. Comparison to the known T_p and T_w aims to identify any defective spectral filters.

3.3 Control and Uncertainties

For the first part of the experiment, the signal, S, is directly measured from the detector. The uncertainty in the signal, S, measurements are calculated from the NER relations presented in Appendix A.4. The uncertainties in the final $\Delta\lambda_f$ values are calculated from the propagation of the uncertainties of the S measurements through Equations (5) and (6). During the experiment and the uncertainty propagation, the blackbody source temperatures T_h and T_c are taken as control variables. Uncertainties in the radiance, $R_f(T)$, are propagated from the uncertainties of the measured $\Delta\lambda_f$'s, through Equation (1). These are represented as shaded areas in the second row of Figure 3. Variables in this part of the experiment are classified in the first row of Table 1

In the second part of the experiment, again a signal, S, is directly measured. And again, the uncertainty in the signal, S, measurements are calculated from the NER relations presented in Appendix A.4. As detailed in Appendix A.3, S is proportional to the input radiance, R_f. Uncertainties in the final T_p and T_w values are propagated from the uncertainties in the signal, S, through Equation (2). These uncertainties are represented as shaded areas in Figure 4. Variables in this part of the experiment are classified in the second row of Table 1

Experiment	**Control Variables**	**Measured Variables**	**Calculated Variables**
First Experiment	T_h, T_w	S, ΔS_w, ΔS_f	$\Delta\lambda_f$, $R_f(T)$, $\tau_f(\lambda)$
Second Experiment	$\Delta\lambda_f$, $\tau_f(\lambda)$	S, ΔS_w, ΔS_f	T_h, T_w

Table 1: Summary of control, measured, and calculated variables in both parts of the experiment. Recall the transmission function, $\tau_f(\lambda)$, from Equation (1).

4 Results

4.1 Harmonic Contribution

The $\Delta\lambda_f$ values obtained in the first part of this experiment following Section 3.2 are presented in Table 2. The first row of Figure 3 presents the two transmission functions obtained for each spectral filter for the cases of with and without harmonic filtering through a thicker ZnSe window. Also, the Planck function is presented over the same domain. As justified in Section 3.2, these transmission functions are single Gaussian curves centred at $\lambda = \lambda_f$ of the respective spectral filter with their FWHM being the respective $\Delta\lambda_f$ (from Table 2).

The second row of Figure 3 presents the solution of Equation (1) for $R_f(T)$ using the Planck function and the respective transmission function from the first row. The temperature domain is selected to be a range containing the room temperature and the temperatures of the blackbody sources used.

Spectral filter	$\Delta\lambda_f$ **without ZnSe window**	$\Delta\lambda_f$ **with ZnSe window**
$\lambda_f = 8.16\mu$m	$0.382 \pm 0.011\mu$m	$0.355 \pm 0.023\mu$m
$\lambda_f = 10.62\mu$m	$0.127 \pm 0.013\mu$m	$0.222 \pm 0.025\mu$m
$\lambda_f = 11.87\mu$m	$0.509 \pm 0.021\mu$m	$0.356 \pm 0.036\mu$m

Table 2: Measurements of $\Delta\lambda_f$ for the spectral filters tested in section 4.1.

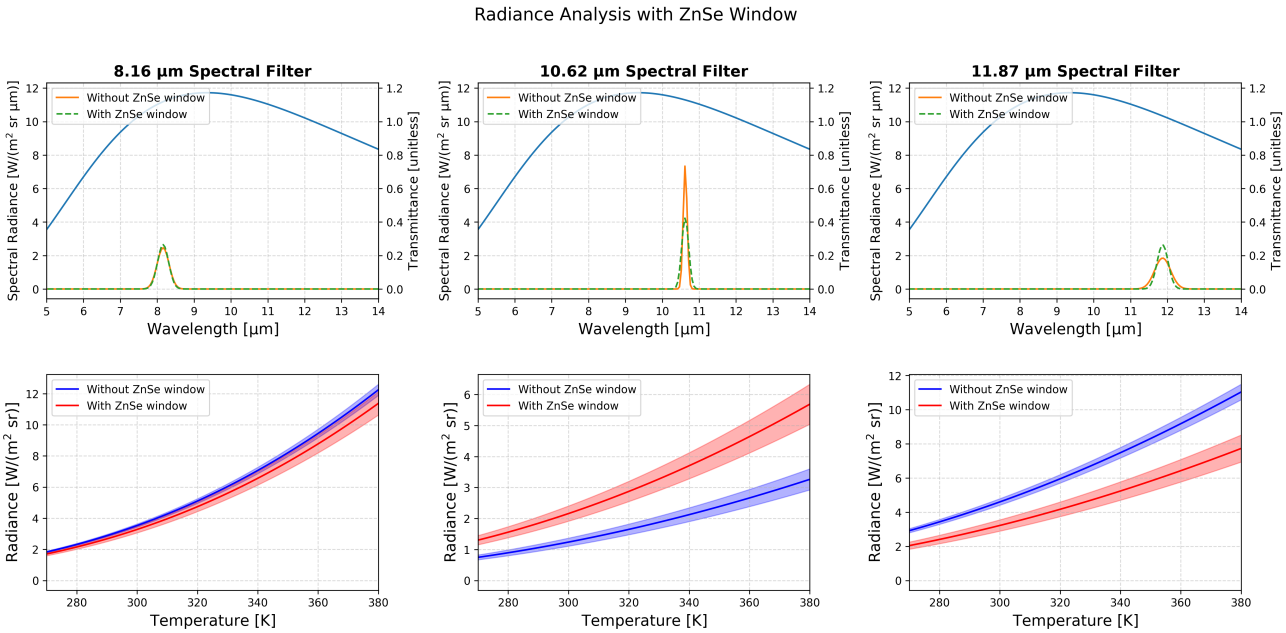


Figure 3: First Row: Transmission functions of the respective spectral filters plotted as a function of wavelength in the cases of with and without a thicker ZnSe window. The blue line is Planck's function. Second Row: Graph of the radiance, $R_f(T)$, calculated using the Planck's function and $\Delta\lambda_f$ from the first row, through Equation (1).

4.2 Parallel Measurements

Table 3 presents the T_p and T_w values obtained from the second part of the experiment. The last row gives the T_p and T_w values directly measured from a high-accuracy thermistor, these are used as the expectation values. As explained in Section 3.2, each remote sensing measurement was done with a pair of spectral filters out of the three spectral filters from the first part of the experiment. The rest of the rows of Table 3 present the remote temperature sensing measurements of the T_p and T_w values and their uncertainties. All measured T_p values lie within 6.62K and all measured T_w values lie within 1.54K of the expected temperatures of 313.3K and 300K, respectively. Figure 4 graphically represents the data from Table 3, plotting the found T_p and T_w values with their uncertainties as shaded areas to highlight the degree of overlap with the expectation values.

Spectral filter pair	**Measured T_p / K**	**Measured T_w / K**
$\lambda_f = 8.16\mu$m & 10.62μm	306.68 ± 1.23	301.54 ± 1.23
$\lambda_f = 10.62\mu$m & 11.87μm	311.97 ± 2.45	299.21 ± 2.45
$\lambda_f = 8.16\mu$m & 11.87μm	306.94 ± 1.78	299.34 ± 1.78
Thermistor Expectation	313.3 ± 0.1	300 ± 0.1

Table 3: Measurements of T_p and T_w with their uncertainties, along with the expected thermistor values.

5 Discussion

5.1 Harmonic Contribution

For the $\lambda_f = 8.16\mu$m spectral filter, the comparison of the input radiances, $R_f(T)$, between the with and without the thicker ZnSe window cases from Figure 3 evidences a non-negligible contribution from the harmonics of λ_f. The $R_f(T)$ lines have their means lying outside each other's error bars. For the spectral filter with $\lambda_f = 11.87\mu$m, this deviation in $R_f(T)$ through the removal of harmonics is even more significant. As seen from Table 2, the change in $\Delta\lambda_f$ between the cases with and without the thicker ZnSe window is significantly larger than the uncertainty for the $\lambda_f = 11.87\mu$m spectral filter.

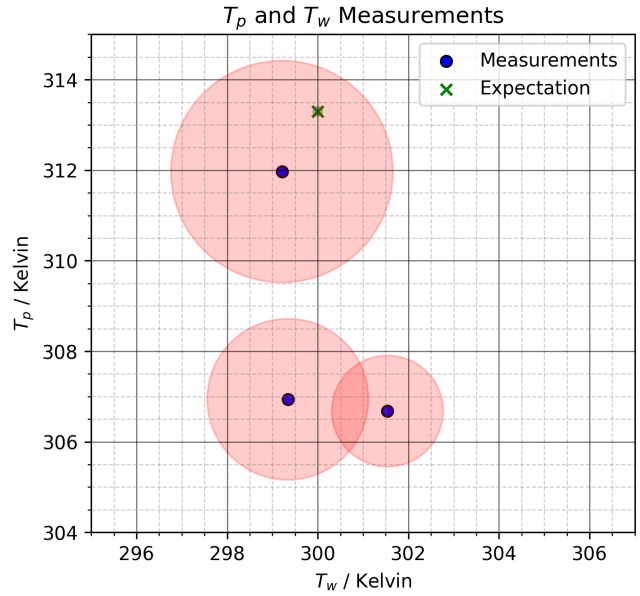


Figure 4: Measured T_p and T_w values plotted with their uncertainties, compared to a high-accuracy thermistor measurement.

In Table 2, for the spectral filter with $\lambda_f = 10.62\mu$m, a larger $\Delta\lambda_f$ is observed for the case where the thicker ZnSe window filters out contributing wavelengths. However, the theory suggests that the contribution from λ_f is now more predominant as the harmonics are filtered out. As a result, a narrower $\tau(\lambda)$ is expected, which implies that the findings for the spectral filter of $\lambda_f = 10.62\mu$m include an anomaly. This claim is supported by the input radiance, $R_f(T)$, curves of this spectral filter in the second row of Figure 3. The curve for the thicker ZnSe window case lies above the curve for the normal ZnSe window case. However, as discussed in Section 3.2, this contradicts the theoretical expectation for a non-anomalous result. Given the increase in $\Delta\lambda_f$ is $\approx 74.8\%$ and the means of the $R_f(T)$ curves lie well outside each others error bars, this anomaly strongly suggests a significant defect in the $\lambda_f = 10.62\mu$m spectral filter. Comparing the $\Delta\lambda_f$ values measured for the $\lambda_f = 10.62\mu m$ spectral filter to the literature findings [9], there is an agreement with the "With ZnSe window" result to 5%. Therefore, the "Without ZnSe window" measurement should be marked as an anomaly.

5.2 Parallel Measurements

Analysis of Figure 2 shows that the transmittance of the CaF_2 window is significantly lower around 11.87μm relative to its transmittance in the 5–10μm range. This leads to a significantly larger relative uncertainty for measurements through the spectral filter with $\lambda_f = 11.87\mu$m. Therefore, as seen in Figure 4, the solutions involving this Spectral Filter have larger uncertainty.

Overall, all of these measurements show strong agreement with the expectation values, where the T_p results lie within 2% of the expected value and the T_w results lie within 0.5% of the expected value. The measured T_p values had a standard deviation of 3.4K, and the measured T_w values had a standard deviation of 1.1K. The close alignment between the measurements and the expectation values while obtaining a low spread relative to the magnitude of the measured values (1.1% and 0.4% for T_p and T_w respectively) allows the justification of confidence in measurements with our three spectral filters, with a remark on possible additional error from the $\lambda_f = 11.87\mu$m spectral filter.

6 Conclusions

The first part of the experiment, showed evidence for a non-negligible contribution from the harmonics of the mean wavelength, λ_f, to the transmittance functions of spectral filters in remote temperature sensing experiments. In two of the three spectral filters used, the results concluded that due to the harmonic contributions, there was a significant increase in the FWHM, $\Delta\lambda_f$, where the third spectral filter was marked as an anomaly. The findings, in agreement with the literature, have concluded that a decrease in $\Delta\lambda_f$ is present when the harmonic contributions are filtered out.

The second part of the experiment conducted three further remote temperature sensing measurements to test the reliability of the equipment used and to identify any defective equipment. All measurements lied within 2% and 0.5% of the expected values, demonstrating high accuracy and precision. This confirmed confidence in the experimental setup, with a remark that the low transmittance of the CaF_2 window (the analogue of an atmosphere in the model) at certain wavelengths could introduce additional errors to one of the spectral filters measurements.

Overall, the results from this experiment suggest that the accuracy of remote temperature sensing experiments which use spectral filters can be improved by filtering out harmonic contributions. This could be done using either a detector that can use Fourier decomposition to separate out the harmonics or an additional optical equipment (the thicker ZnSe window in our case) to physically filter out the harmonics. The latter approach comes with an increase in optical path and a decrease in the input radiance, which may not be well applicable to low-intensity beams. In industrial applications, these findings particularly have implications on laser-based sensing systems (e.g. LIDAR) where harmonic-induced spectral broadening reduces distance measurement precision.

References

[1] S. Khorram. *Remote sensing*. eng. First edition 2012. SpringerBriefs in space development. New York: Springer, 2012. ISBN: 1-280-79381-3.

[2] M. Bevis et al. "GPS meteorology: Remote sensing of atmospheric water vapor using the global positioning system". eng. In: *Journal of Geophysical Research* 97.D14 (1992), pp. 15787–15801. ISSN: 0148-0227.

[3] D. L. Tang et al. "Remote sensing oceanography of a harmful algal bloom off the coast of southeastern Vietnam". eng. In: *Journal of Geophysical Research. C. Oceans* 109.C3 (2004), C03014.1–n/a. ISSN: 0148-0227.

[4] I. S. Robinson. *Satellite oceanography : an introduction for oceanographers and remote-sensing scientists*. eng. Ellis Horwood series in marine science. Chichester, West Sussex, England: E. Horwood, 1985. ISBN: 0853125988.

[5] V. V. Busarev. "Spectrophotometry and Other Remote-Sensing Methods to Study Asteroids: Achievements and New Approaches". eng. In: *Solar system research* 57.1 (2023), pp. 61–75. ISSN: 0038-0946.

[6] A. Dudhia. *Remote Sounding*. The University of Oxford, 2023. URL: `https://www-teaching.physics.ox.ac.uk/practical_course/scripts/srv/local/rscripts/trunk/Atmos/AP01/AP01.pdf`. (accessed: 13.12.2024).

[7] S. Hansen et al. "SOFIE Instrument Ground Calibration". In: *Proceedings of SPIE* (2006). DOI: `10.1117/12.681164`.

[8] T. Matsumoto et al. "A search for the near-infrared extragalactic background light". In: *Astrophysical Journal* (1988). DOI: `10.1086/166678`.

[9] G. C. Sloan et al. "INFRARED SPECTRAL PROPERTIES OF M GIANTS". In: *The Astrophysical Journal* (2015). DOI: `10.1088/0004-637X/811/1/45`.

[10] O. Hagolle et al. "The Physics of Optical Remote Sensing". eng. In: *Remote Sensing Imagery*. Hoboken, USA, 2014, pp. 53–81. ISBN: 9781848215085.

[11] NEB. *Radiometer experiment mini-project*. The University of Oxford, 2016. URL: `https://www-teaching.physics.ox.ac.uk/practical_course/scripts/srv/local/rscripts/trunk/Atmos/AP01/AP01_mp.pdf`. (accessed: 28.02.2025).

Appendix

A Background Theory

A.1 The Planck Function

The spectral radiance of a blackbody at temperature T is given by the Planck function [6]

$$B(\lambda, T) = \left(\frac{2hc^2}{\lambda^5}\right) \frac{1}{\exp\left[hc/(\lambda kT)\right] - 1} \ , \tag{7}$$

with the units of [W/(m^2 sr μm)]. Plotted at Figure 5 for different temperatures near the room temperature.

The total radiance of a blackbody is defined as

$$\int_0^\infty B(\lambda, T)\, d\lambda = \left(\frac{2\pi^4 k^4 T^4}{15 h^3 c^2}\right) \equiv \bar{B}(T) \ , \tag{8}$$

with the units of [W/(m^2 sr)]. Where h (Planck's constant), c (speed of light), and k (Boltzmann constant) are fundamental physical constants.

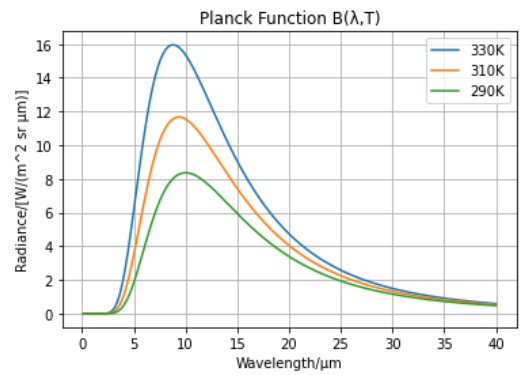


Figure 5: Planck Function plotted at three different temperatures.

The Stefan-Boltzmann law models the intensity of radiation emitted from a blackbody as a function of its temperature. It can be derived by integrating the radiance emitted per unit area over a hemisphere [6]

$$\int_0^{2\pi} d\phi \int_0^{\pi/2} \bar{B}(T) \cos\theta \sin\theta \ d\theta = \pi \bar{B}(T) = \left(\frac{2\pi^5 k^4 T^4}{15 h^3 c^2}\right) \equiv \ \sigma T^4 \ , \tag{9}$$

with the units of [W/m^2]. The $\cos\theta$ factor accounts for the projection of the area depending on the viewing angle θ from the normal.

A.2 Etendue

In an optical system such as in Figure 6, an image of the target falls onto the detector. And from the principle of reversibility for light, an image of the detector also falls on the target [6].

Etendue is a conserved quantity when the target and the detector are exchanged, given by $A\Omega$ [10], where A is the cross-sectional area of the aperture and Ω is the solid angle subtended by the light beam.

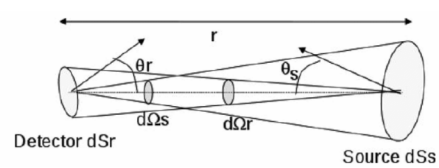


Figure 6: Descriptive scheme for defining the geometric etendue. Reproduced from [10].

Conservation of etendue can be demonstrated as

$$ds_r d\Omega_s \cos\theta_r = ds_r \frac{ds_s \cos\theta_s}{r^2} \cos\theta_r = ds_s \frac{ds_r \cos\theta_r}{r^2} \cos\theta_s = ds_s d\Omega_r \cos\theta_s \ , \tag{10}$$

where the quantities are presented in Figure 6.

A.3 Radiometer Signal

This practical configures the radiometer and the electronics such that the signal obtained is proportional to the input radiance. Further discussion can be found in [11].

The radiative power, P, collected by a radiometer is given by [6]

$$P = A\Omega \int R(\lambda)\, \tau(\lambda)\, d\lambda \ , \tag{11}$$

where $A\Omega$ is the etendue from viewing a solid angle Ω by a primary aperture of area A. $R(\lambda)$ is the spectral radiance passing through optical components of net transmittance $\tau(\lambda) = \prod_i \tau_i(\lambda)$. The transmittance of the spectral filters will limit the radiance to a small domain in wavelength close to λ_f.

The radiative power collected is integrated over time to get a voltage signal which, as stated, is proportional to the input radiance.

A.4 Noise Analysis

Each detector has a characteristic Noise Equivalent Power (NEP) in the units of [W $\mathrm{Hz}^{-1/2}$]. This quantity represents the detector noise which is averaged out to a certain degree, with respect to the integration time – hence the frequency dependence [6].

Conversion of this measurement error into an uncertainty in the radiance measurement results in the Noise Equivalent Radiance (NER). Imposing a small wavelength domain approximation due to the spectral filters we can evaluate Equation (11) as

$$P = A\Omega \int R\left(\lambda\right) \tau_f\left(\lambda\right) d\lambda \simeq A\Omega R\left(\lambda_f\right) \tau_0 \Delta\lambda_f \ . \tag{12}$$

The integration time over which the detector accumulates signal causes a factor of $\sqrt{\Delta t}$, with NEP and NER related as

$$\mathrm{NER} = \frac{\mathrm{NEP}}{A\Omega\tau_0\Delta\lambda_f\sqrt{\Delta t}} \ . \tag{13}$$

The main limitation of instrumentation for NER is that a larger detector, so a larger etendue, would improve NER, but a larger spectral filter that accommodates for a larger detector could also result in a larger $\Delta\lambda_f$, counteracting this improvement.